

Chapter 3 — Vector Foundations

3.1 Transforming Information into Geometry

Provenance: NEWLY DEVELOPED FROM VALIDATED NOTEBOOK

Canonical companion: [M1_N03_vector_foundations.ipynb](#)

Once information becomes a vector, questions about similarity, direction, magnitude and projection become mathematical questions.

3.2 Learning Objectives

The reader will learn to distinguish scalars, points and vectors; calculate vector operations and norms; interpret distance, dot products and cosine similarity; compute projections; understand span and orthogonality; construct an orthonormal basis; and represent observations, embeddings and decision states as vectors.

3.3 Opening Story — Two Districts That Look Similar

Two agricultural districts have similar total cereal production. A quick comparison suggests they face the same policy situation.

Yet one district has high rainfall, low fertilizer use and limited market access. The other has lower rainfall, intensive inputs and strong market connections.

A single number hides their different structures.

Represent each district by a vector:

$$x = \begin{bmatrix} \text{rainfall} \\ \text{fertilizer} \\ \text{yield} \\ \text{market access} \end{bmatrix}.$$

The districts can now be compared as profiles rather than totals. But the comparison remains meaningful only if units, scaling and indicator direction are treated correctly.

3.4 What Is a Vector?

A scalar records magnitude:

$$x = 5.$$

A vector records ordered components:

$$x = (x_1, \dots, x_n)^T.$$

Depending on context, a vector may represent a displacement, an observation, a feature profile, a probability distribution, an embedding, a gradient or the state of a digital twin.

Vector addition and scalar multiplication are defined component by component:

$$x + y = (x_1 + y_1, \dots, x_n + y_n),$$
$$c x = (c x_1, \dots, c x_n).$$

These simple operations allow complex states to be combined and adjusted.

3.5 Magnitude and Norms

Different norms express different ideas of size:

$$\|x\|_1 = \sum_i |x_i|, \|x\|_2 = \sqrt{\sum_i x_i^2}, \|x\|_\infty = \max_i |x_i|.$$

The L^1 norm accumulates absolute magnitudes. The L^2 norm gives Euclidean length. The L^∞ norm focuses on the largest absolute component.

No norm is universally correct. The appropriate choice depends on what distance or error means in the problem.

Euclidean distance is

$$d(x, y) = \|x - y\|_2.$$

If indicators use incompatible units, raw Euclidean distance may be dominated by the variable with the largest numerical scale.

3.6 Dot Product, Angle and Similarity

The dot product is

$$x^\top y = \sum_i x_i y_i.$$

For nonzero vectors,

$$\cos \theta = \frac{x^\top y}{\|x\|_2 \|y\|_2}.$$

Orthogonal vectors have zero dot product. For nonzero vectors, cosine similarity equals 1 for the same direction, 0 for orthogonal directions and -1 for opposite directions. Cosine similarity is undefined if either vector is zero.

Cosine similarity emphasizes orientation rather than magnitude. This makes it useful for text embeddings, but it does not prove semantic equivalence, causation or policy interchangeability.

3.7 Projection

The projection of vector x onto a nonzero direction u is

$$\text{proj}_u(x) = \frac{x^\top u}{u^\top u} u.$$

The rejection

$$x - \text{proj}_u(x)$$

is orthogonal to u ; equivalently, $u^T r = 0$.

Projection separates the component aligned with a chosen direction from the component that lies outside it. This idea later becomes central to regression, principal-component analysis and attention mechanisms.

3.8 Linear Combinations, Span and Bases

A linear combination is

$$c_1 v_1 + \dots + c_k v_k.$$

The set of every such combination is the span. A basis is a linearly independent set that spans a space.

For linearly independent vectors $v_1, \dots, v_k$, Gram-Schmidt constructs an orthonormal basis by removing components along directions already constructed:

$$q_1 = \frac{v_1}{\|v_1\|_2}.$$
$$u_k = v_k - \sum_{j=1}^{k-1} (v_k^T q_j) q_j.$$
$$q_k = \frac{u_k}{\|u_k\|_2}.$$

For example, applying the procedure to $v_1 = (1, 0)^T$ and $v_2 = (1, 1)^T$ gives the standard orthonormal basis:

$$v_1 = (1, 0)^T, v_2 = (1, 1)^T \Rightarrow q_1 = (1, 0)^T, q_2 = (0, 1)^T.$$

An orthonormal basis simplifies coordinates because dot products recover component weights directly.

3.9 Statistics and AI

For n observations and p variables,

$$X \in \mathbb{R}^{n \times p}$$

contains observation vectors as rows and variable vectors as columns.

AI represents engineered features, tokens, images, hidden states, gradients, attention queries, keys and values as vectors. Much of modern AI is therefore high-dimensional geometry.

3.10 Decision Intelligence Case — National State Vectors

A national state vector might include standardized indicators for growth, inflation, food production, electricity access and health capacity.

Before comparison:

- indicators must be scaled;
- undesirable indicators may need their direction reversed;
- weights must be justified;
- uncertainty should be recorded;
- missing values must not be silently replaced.

Geometric similarity describes similarity under the chosen representation. It does not establish equal welfare or identical policy needs.

3.11 AI Reasoning Clinic

3.11.1 Statement

“These two documents have cosine similarity 0.94, so they mean the same thing.”

3.11.2 Evaluation

The conclusion is too strong. The value describes similarity between embeddings under a particular model. It may be affected by tokenization, model training, text length and domain. Human or task-specific validation remains necessary.

3.12 Computational Checkpoint

Before accepting a vector calculation, verify all of the following:

1. Every vector has the declared dimension and coordinate order.
2. Norms, distances and similarities use explicit definitions and valid domains.
3. Projection and normalization never use a zero direction vector.
4. Scaling, weighting and indicator direction are documented before interpretation.

3.13 Scaling Changes Geometry

Suppose one coordinate is measured in millimetres and another in kilometres. Raw Euclidean distance can then be driven mainly by the numerical scale of the first coordinate rather than by substantive difference. Standardization changes the geometry by expressing coordinates on comparable scales; weighting changes it again by assigning different importance to directions. Neither choice is neutral, so both must be recorded and tested.

3.14 Numerical Safeguards

Vector addition and the dot product require compatible dimensions. Cosine similarity requires two nonzero vectors. Projection requires a nonzero direction. Gram–Schmidt requires independent input directions if the requested output must retain the same dimension.

Numerical implementations should use tolerances for floating-point comparisons, test orthogonality through residual dot products, and report how normalization, standardization or weighting changes results. A geometric output is meaningful only under the representation that produced it.

3.14.1 Verification Questions

Are dimensions and coordinate order compatible? Are any denominators zero or numerically negligible? Which units, scaling rule and weights define the geometry? Do independent calculations confirm the norm, projection or orthogonality claim? Would a small perturbation change the substantive interpretation?

3.15 Worked Mathematical Example

Let $x=(3,4)^T$ and let $u=(1,0)^T$. Because u is nonzero, the projection is defined. Compute the aligned component, the rejection and an independent orthogonality check:

$$x=(3,4)^T, u=(1,0)^T.$$

$$proj_u(x) = \frac{x^T u}{u^T u} u = 3(1,0)^T = (3,0)^T.$$

$$r = x - proj_u(x) = (0,4)^T.$$

$$u^T r = (1,0)(0,4)^T = 0.$$

A small perturbation of the second coordinate leaves the projection unchanged but changes the rejection magnitude:

$$x'=(3,4.1)^T \Rightarrow proj_u(x')=(3,0)^T, \|r'\|_2=4.1.$$

The paper derivation agrees with the notebook functions `project` and `reject`. The residual check $u^T r=0$ verifies the defining property directly. The perturbation shows that projection onto u depends only on the component aligned with u , while the orthogonal component records the change.

3.16 Professional Applications

Official statistics uses these foundations in sampling, weighting, estimation, variance, disclosure control and reproducible publication. Agricultural statistics uses them for indicators, forecasting, geospatial analysis and uncertainty. Machine learning uses them for representations, loss functions, gradients, probabilities and optimization. Policymakers need conclusions translated into magnitude, uncertainty, assumptions and decision consequence.

3.17 Verification Clinic

Claim: “The software returned a precise vector result, so the calculation is correct.”

Evaluation: Display precision is not mathematical validity. Dimensions, domains, units, zero-vector restrictions, scaling choices and numerical error remain to be checked.

Improved reasoning: Restate the objects and assumptions, reproduce a small case independently, test invariants such as $u^T r = 0$ or $Q^T Q = I$, perturb the inputs, and communicate which conclusions depend on representation choices.

3.18 Failure Modes and Safeguards

- Undefined objects, mismatched dimensions or incompatible units.
- Use of a theorem or distribution outside its stated conditions.
- Numerically unstable formulas that amplify rounding or conditioning.
- Interpretation of mathematical output as causal, certain or self-authorizing.

Safeguards include explicit definitions, symbolic derivation, boundary cases, dimensional analysis, reference examples, stable algorithms, residual and convergence checks, reproducible computation, sensitivity analysis and professional review.

3.19 Notebook Verification Checklist

1. Run every cell from a clean kernel and retain the execution evidence.
2. Confirm expected failures for normalization, cosine similarity and projection involving zero vectors.
3. Check projection rejection directly with $u^T r = 0$ and Gram–Schmidt with $Q^T Q = I$.
4. Compare raw, standardized and reweighted state vectors before interpreting similarity.

3.20 Author’s Perspective

Mathematics is the language that makes AI reasoning examinable. It does not remove judgment; it disciplines judgment by revealing definitions, structure, uncertainty and consequence. Reproducible computation extends that discipline from the page to the executable system.

3.21 Chapter Summary and Key Insight

Vectors represent ordered quantities in coordinate systems. Addition, scaling, inner products, norms, angles and distance provide the geometry underlying features, similarity, gradients and optimization. The canonical notebook turns the reasoning into an inspectable experiment, while professional use still requires valid assumptions, stable computation, uncertainty communication and accountable interpretation.

3.22 SRAI Principle No. 3

Geometry becomes evidence only after representation, scaling and interpretation have been justified.

3.23 Exercises and Laboratory

1. Compute the L^1 , L^2 and L^∞ norms for the same vector.
2. Prove that the projection rejection is orthogonal to the nonzero projection direction u .
3. Compare Euclidean distance before and after standardization.
4. Implement cosine similarity without a library function.
5. Use M1_N03 to create a multi-sector state vector and document how changing one scaling choice alters the conclusion.

3.24 Key Insight and Reflection

Vectors transform information into geometry. That transformation enables comparison, projection, optimization and learning. It also introduces modeling choices that must remain visible.

3.25 Looking Ahead

Chapter 4 organizes vectors into matrices. Matrix algebra enables us to represent entire datasets and systems of relationships, then solve the linear systems that arise throughout statistics and AI.

3.26 Verified References and Further Reading

- NumPy documentation. Linear algebra reference.
<https://numpy.org/doc/stable/reference/routines.linalg.html>
- NumPy documentation. Vector and matrix norms.
<https://numpy.org/doc/stable/reference/generated/numpy.linalg.norm.html>

Technical audit note: Definitions, formulas and computational claims were checked against the cited primary documentation and independently executed reference calculations. Final publication remains subject to owner approval.

3.27 Student Assessment - Chapter 3

1. Compute the dot product of (1,2) and (3,4).
2. Compute the Euclidean norm of (3,4).
3. When are two nonzero vectors orthogonal?
4. Why must features be scaled before distance methods?
5. What does cosine similarity omit?

3.28 Instructor Solutions - Chapter 3

1. **Answer:** 11.

Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.

2. **Answer:** 5.

Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.

3. **Answer:** Their inner product is zero.

Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.

4. **Answer:** Large-unit coordinates can dominate geometric distance.

Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.

5. **Answer:** It compares direction and ignores magnitude, subject to zero-vector handling.

Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.